Notebook Guide

> Purpose, scope, and recommended usage for the Jupyter notebooks in `notebooks/`.

Overview

The eight notebooks in this project are **exploratory and analytical companions** to the production pipeline, not standalone end-to-end runners. They were created during development to prototype analysis, visualize intermediate results, and document findings for each project phase. The production workflow is driven by the scripts in `scripts/` and the `src/ems_readiness` package.

Bottom line: Use `scripts/` to reproduce results. Use `notebooks/` to understand, explore, and present them.

Notebook Inventory

Notebook	Phase	Purpose	Standalone?
`02_eda_spatiotemporal`	1 — EDA	Spatiotemporal analysis of crash demand patterns; generates hourly/daily/seasonal figures	Yes (reads raw data directly)
`03_input_modeling`	2 — Demand	Documents NHPP demand model fitting, lambda tables, goodness-of-fit	Partially (reads processed data)
`04_service_travel_proxy`	2 — Service	Demonstrates distance matrix, travel-time proxy, service-time distribution, arrival generator	No (imports `ems_readiness` package)
`05_optimization`	3 — Optimization	Documents optimization formulations and policy comparison results	Partially (reads results CSVs)
`06_simulation_debug`	4 — Simulation	Verification and validation of the DES engine (toy examples, pilots)	No (imports `ems_readiness` package)
`07_production_results`	5 — Results	Analyzes 1,440 production simulation runs across four experiment sets	Partially (reads results CSVs)
`08_statistical_analysis`	6 — Statistics	Full statistical analysis: ANOVA, Tukey HSD, effect sizes, confidence intervals	Partially (reads results CSVs)
`09_cbd_analysis`	Robustness	CBD-focused optimization and distance-metric comparison	Partially (reads results CSVs)

"Standalone?" column explained

- **Yes**: The notebook reads raw data and can run independently given the raw data files.

- **Partially**: The notebook reads pre-computed results (CSVs from `data/processed/` or `results/`). It can run if those files exist, but does not regenerate them — that is the job of `scripts/`.
- **No**: The notebook imports `ems_readiness` as a library (via `sys.path` injection) and depends on the installed or path-accessible package.

Relationship to `scripts/`

```
scripts/                          notebooks/
──────────────────────────────    ──────────────────────────────
data_audit.py                  -> 02_eda_spatiotemporal.ipynb
demand_modeling.py             -> 03_input_modeling.ipynb
(package modules)              -> 04_service_travel_proxy.ipynb
run_optimization_comparison.py -> 05_optimization.ipynb
run_verification.py            -> 06_simulation_debug.ipynb
run_validation_pilots.py
run_production_experiments.py  -> 07_production_results.ipynb
run_production_v2.py
analyze_production_results.py  -> 08_statistical_analysis.ipynb
run_cbd_experiment.py          -> 09_cbd_analysis.ipynb
run_cbd_focused_optimization.py
```

Key points:

- Scripts generate data and results; notebooks visualize and analyze them.
- Scripts are the authoritative, reproducible pipeline. Notebooks may contain exploratory code that was later formalized into scripts.
- Four notebooks (`05`, `06`, `07`, `08`) have paired `.py` files in `notebooks/` — these are Jupytext percent-format scripts used for version control of the notebook content.
- No notebook calls a script via `subprocess`. The `08_statistical_analysis` notebook references script paths in documentation but loads results files directly.

Recommended Workflow

To reproduce all results from scratch

```
# 1. Process raw data
make data

# 2. Run the full analysis pipeline
make analysis

# 3. Or run individual scripts
python scripts/run_optimization_comparison.py
python scripts/run_production_v2.py
python scripts/analyze_production_results.py
```

To explore or present results

Open notebooks in JupyterLab after running the production scripts:

```
jupyter lab notebooks/
```

Notebooks assume results files already exist in `data/processed/` and `results/`.

To modify the simulation or optimization

1. Edit source code in `src/ems_readiness/`
2. Re-run the appropriate script in `scripts/`
3. Open the corresponding notebook to visualize updated results

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Jupytext `.py` Files

The `.py` files in `notebooks/` (e.g., `05_optimization.py`) are **Jupytext percent-format** mirrors of the corresponding `.ipynb` files. They exist for cleaner version control (diffs on `.py` are more readable than on `.ipynb` JSON). These are not independent scripts — they are an alternative representation of the same notebook content.

Maintenance Status

All notebooks reflect the final project state (V2 with spatially-stratified P0, capacity=2 default, CBD robustness analysis). They are maintained as documentation and presentation artifacts. For any future model changes, update the scripts first, re-run experiments, then refresh notebook outputs.